

EE227 – Digital Logic Design

Lecture 12

Outline

- Exclusive OR (XOR) Function
- Parity Generation and Checking
- Quiz 2 Solution
- Mid Term I Solution and Result

Exclusive OR Function (XOR)

- The Exclusive-OR (XOR) is denoted by $\oplus$ and performs the following logical operation

$$A \oplus B = AB' + A'B$$

- Exclusive NOR is complement of XOR
- The following identities are true for XOR

$$A \oplus 0 = A \quad A \oplus 1 = A' \quad A \oplus A = 0$$

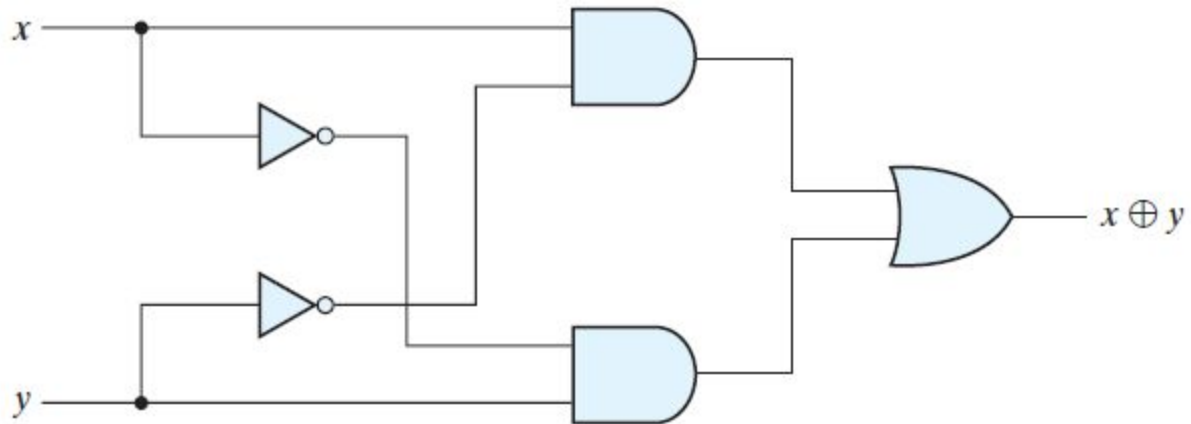
$$A \oplus A' = 1 \quad A \oplus B' = A' \oplus B = (A \oplus B)'$$

- The XOR operation is commutative and Associative

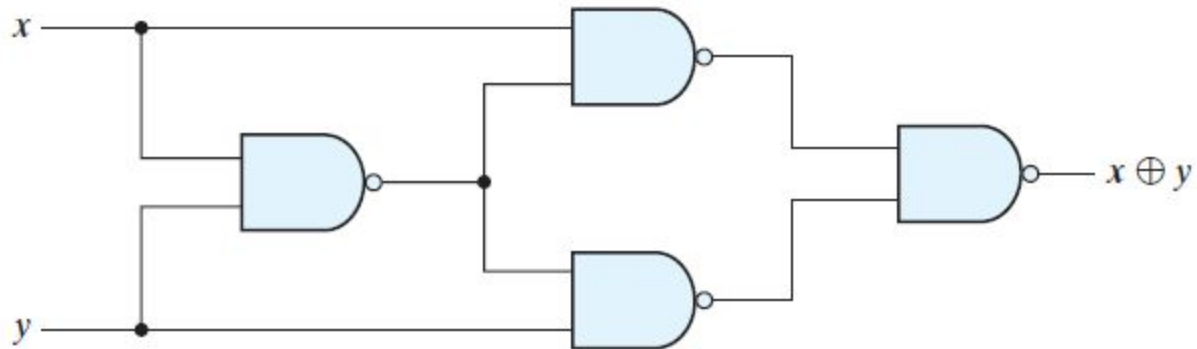
$$A \oplus B = B \oplus A$$

$$(A \oplus B) \oplus C = A \oplus (B \oplus C) = A \oplus B \oplus C$$

XOR with NAND Gates



Exclusive-OR with AND-OR-NOT gates



Exclusive-OR with NAND gates

XOR is an Odd Function

- The output of XOR is equal to 1 if an odd number of inputs are equal to 1
- For 2 inputs

$$A \oplus B = AB' + A'B = \Sigma(1, 2)$$

- For 3 inputs

$$\begin{aligned} A \oplus B \oplus C &= (AB' + A'B)C' + (AB + A'B')C \\ &= AB'C' + A'BC' + ABC + A'B'C \\ &= \Sigma(1, 2, 4, 7) \end{aligned}$$

- For 4 inputs

$$\begin{aligned} A \oplus B \oplus C \oplus D &= (AB' + A'B) \oplus (CD' + C'D) \\ &= (AB' + A'B)(CD + C'D') + (AB + A'B')(CD' + C'D) \\ &= \Sigma(1, 2, 4, 7, 8, 11, 13, 14) \end{aligned}$$

Even Function

- The complement of XOR is an even function

- For 2 inputs

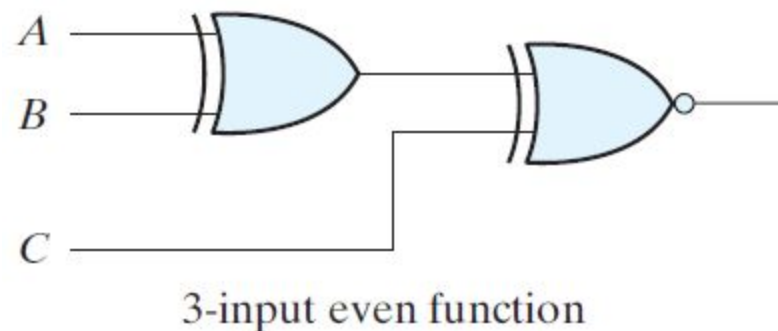
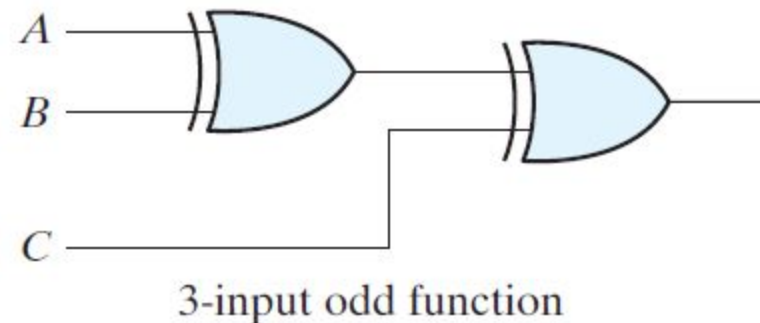
$$(A \oplus B)'$$

- For 3 inputs

$$(A \oplus B \oplus C)'$$

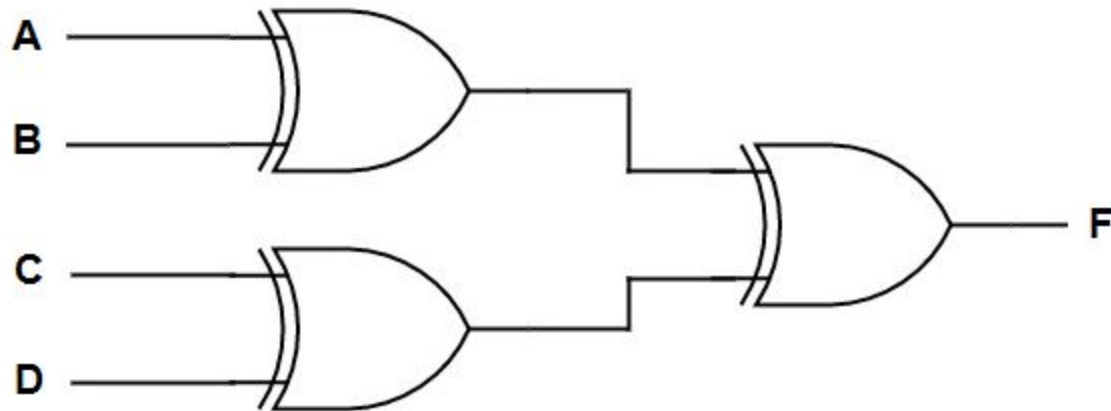
- For 4 inputs

$$(A \oplus B \oplus C \oplus D)'$$



Practice Problem 1

- Construct the circuit diagram for 4 input odd function



Parity Generation and Checking

- Exclusive-OR functions are very useful in systems requiring error detection and correction codes
- This is done with the help of a parity bit
 - The parity bit is an extra bit added to the original message to make the number of 1's even or odd
 - Parity bit is added at MSB position
 - For example the message 011 0010 will be written as
1011 0010 with even parity and
0011 0010 with odd parity
- The circuit that generates the parity bit in the transmitter is called a **parity generator**
- The circuit that checks the parity in the receiver is called a **parity checker**

Parity Generator and Checker (1/3)

- x, y, and z are inputs
- For even parity the number of 1's including Parity bit (P) must be even

$$P = x \oplus y \oplus z$$

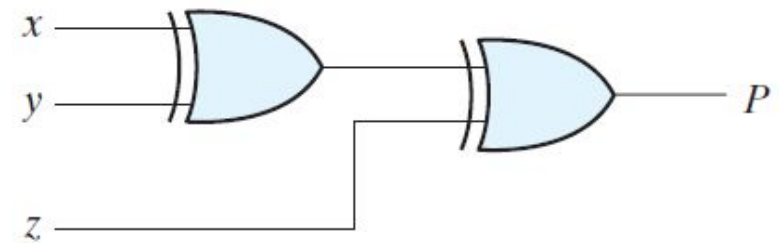
- For parity checker the inputs are x, y, z, and P and the output is 1 if the number of 1's are odd, which means that an error has occurred

$$C = x \oplus y \oplus z \oplus P$$

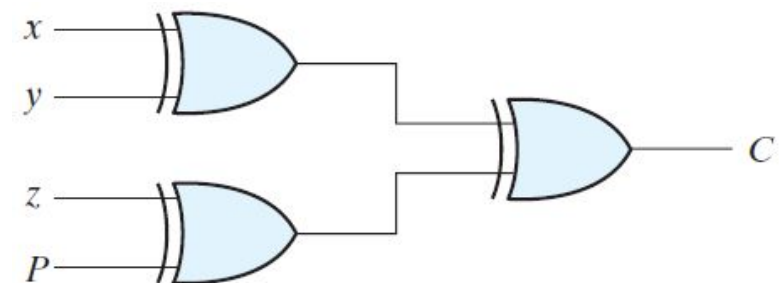
Parity Generator and Checker (2/3)

Even-Parity-Generator Truth Table

Three-Bit Message			Parity Bit
<i>x</i>	<i>y</i>	<i>z</i>	<i>P</i>
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	1



3-bit even parity generator



4-bit even parity checker

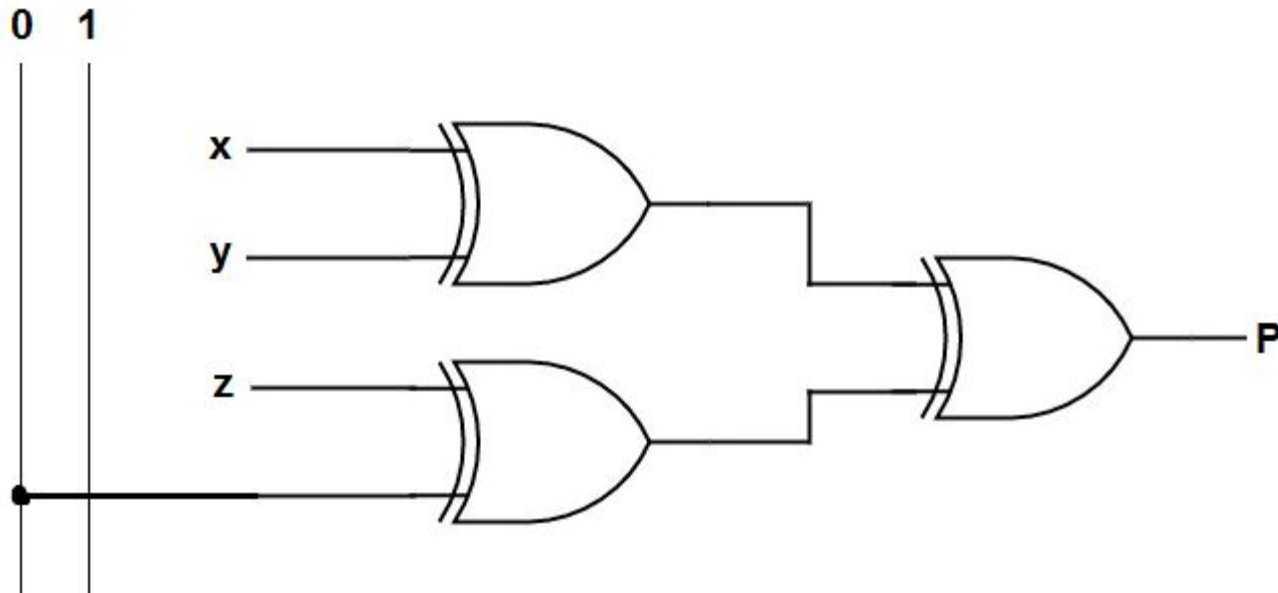
Parity Generator and Checker (3/3)

- The parity generator can be implemented with the circuit of parity checker
- This can be done by connecting the input P with the logic 0 and marking the output as P

Four Bits Received				Parity Error Check
<i>x</i>	<i>y</i>	<i>z</i>	<i>P</i>	<i>C</i>
0	0	0	0	0
0	0	0	1	1
0	0	1	0	1
0	0	1	1	0
0	1	0	0	1
0	1	0	1	0
0	1	1	0	0
0	1	1	1	1
1	0	0	0	1
1	0	0	1	0
1	0	1	0	0
1	0	1	1	1
1	1	0	0	0
1	1	0	1	1
1	1	1	0	1
1	1	1	1	0

Practice Problem 2

- Modify the circuit of 4 bit parity checker to convert it to a 3 bit even parity generator



Quiz 2

Solution

Quiz 2 (Paper A)

Question Number 1

Prove the following Boolean equation using Boolean algebra:

$$xyz' + x'yz' + xy'z' + x'y'z' + z = 1$$

L.H.S

$$\begin{aligned} & xyz' + x'yz' + xy'z' + x'y'z' + z \\ &= xz'(x + x') + x'z'(y + y') + z \\ &= xz' + x'z' + z \\ &= z'(x + x') + z \\ &= z' + z \\ &= 1 \\ &= \text{R.H.S} \end{aligned}$$

Quiz 2 (Paper A)

Question Number 2

Write the Boolean equation (in simplified form) and draw the logic diagram of the circuit whose output is defined by the following truth table.

Hint: Use minterms or maxterms

a	b	c	F
0	0	0	1
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	0

By using minterms

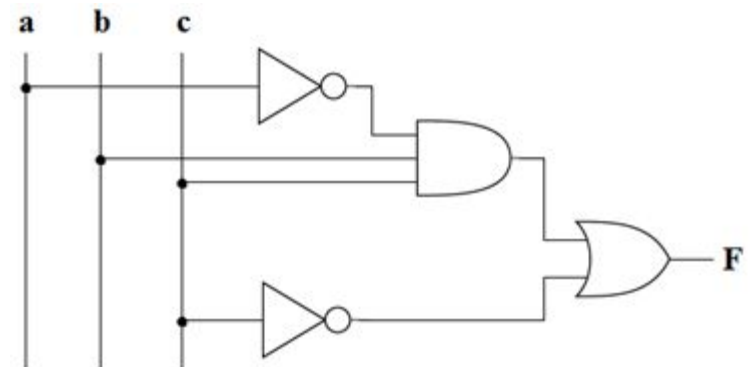
$$F = a'b'c' + a'bc' + a'bc + ab'c' + abc'$$

$$F = a'c'(b' + b) + a'bc + ac'(b' + b)$$

$$F = a'c' + a'bc + ac'$$

$$F = c'(a' + a) + a'bc$$

$$F = c' + a'bc$$



Quiz 2 (Paper B)

Question Number 1

Reduce the following Boolean expression to one literal.

$$A'B(D' + C'D) + B(A + A'CD)$$

$$F = A'B(D' + C'D) + B(A + A'CD)$$

$$F = B(A'D' + A'C'D + A + A'CD)$$

$$F = B(A'D' + A'D(C + C') + A)$$

$$F = B(A'(D' + D) + A)$$

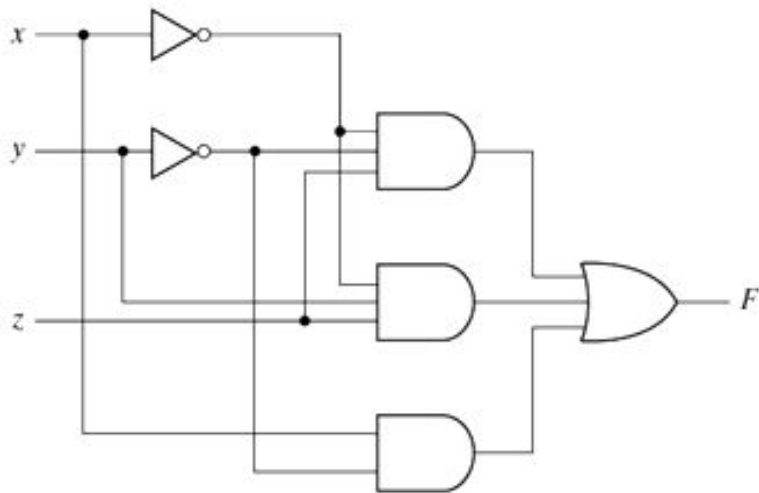
$$F = B(A' + A)$$

$$F = B$$

Quiz 2 (Paper B)

Question Number 2

For the given circuit, write the equivalent Boolean expression and construct the truth table



From circuit the expression is

$$F = x' y' z + x' y z + x y'$$

x	y	z	x'	y'	x'y'z	x'yz	xy'	F
0	0	0	1	1	0	0	0	0
0	0	1	1	1	1	0	0	1
0	1	0	1	0	0	0	0	0
0	1	1	1	0	0	1	0	1
1	0	0	0	1	0	0	1	1
1	0	1	0	1	0	0	1	1
1	1	0	0	0	0	0	0	0
1	1	1	0	0	0	0	0	0

Mid Term I

Solution

Was Done in Class